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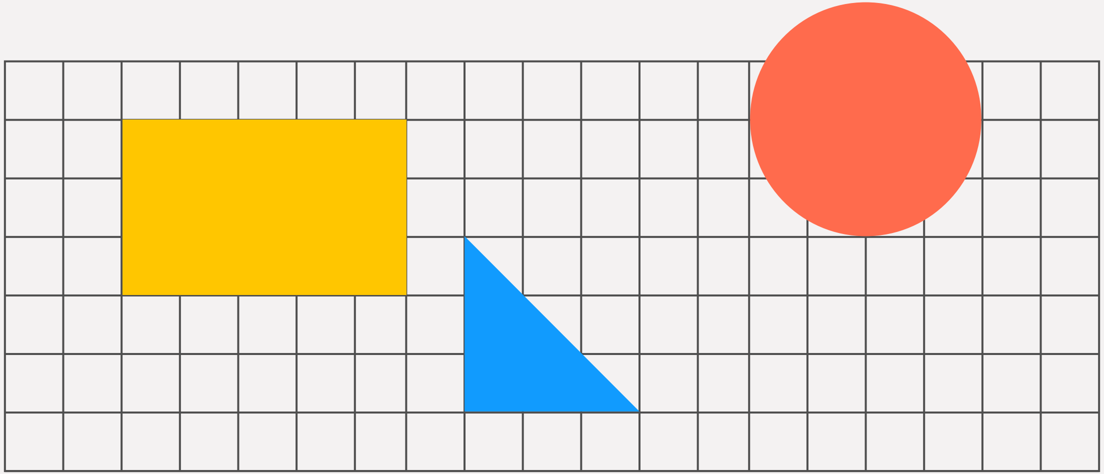
7 May 2026

Student Placement Salary Analysis



Agenda

- Introduction
- Data Cleaning
- Data Analysis
- Conclusion

Introduction

Research question: Which student-related factors best explain salary among students who received job placements?

- Source: Kaggle
- Original dataset: 10,000 observations, 13 variables
- Modeling dataset: 6,153 placed students
- ❖ Removed placement_status from modeling data
- ❖ No missing values / no duplicates in cleaned analysis dataset

Dataset and Cleaning Approach

Original Data → Placed Students Only → Remove placement_status → Final Regression Dataset

Original dataset:
10,000 observations, 13 variables

Steps:
1. Filtered to placement_status = "Placed"
2. Removed placement_status after filtering
3. Checked for missing values and duplicates
4. Converted categorical predictors into factors
5. Final modeling dataset: 6,153 placed students

Why this matters:
Unplaced students had salary = 0, so including them would mix placement prediction with salary prediction.

Response and Predictor Variables

Response Variable

salary

Numeric Predictors

cgpa
backlogs
internship_count
aptitude_score
communication_score
internship_quality_score

Categorical Predictors (factored)

College_tier
Country
University_ranking_band
Specialization
industry

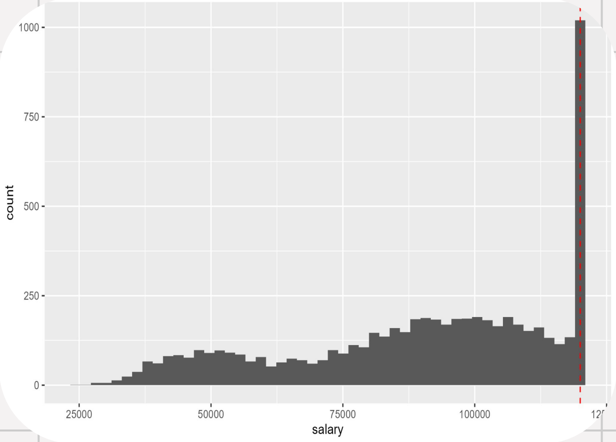
predictor	coefficient	std_error	t_statistic	p_value	r_squared	adj_r_sq	f_statistic	f_p_value
cgpa	3383.703	421.252	8.032	0	0.0104	0.0102	64.521	0
backlogs	-966.14	331.791	-2.912	0.0036	0.0014	0.0012	8.479	0.0036
internship_count	47.667	255.028	0.187	0.8517	0	-2e-4	0.035	0.8517
aptitude_score	-38.634	21.899	-1.764	0.0778	5e-4	3e-4	3.112	0.0778
communication_score	-10.265	21.578	-0.476	0.6343	0	-1e-4	0.226	0.6343
internship_quality_score	40.932	214.484	0.191	0.8487	0	-2e-4	0.036	0.8487

predictor	r_squared	adj_r_sq	f_statistic	df_model	f_p_value
college_tier	0.1967	0.1965	753.111	2	0
country	0.6978	0.6976	3549.087	4	0
university_ranking_band	0.0038	0.0035	11.813	2	0
specialization	0.0304	0.0298	48.193	4	0
industry	0.0074	0.0066	9.145	5	0

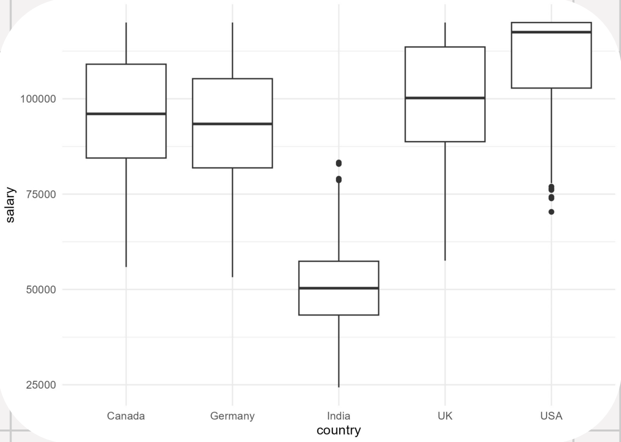
Exploratory Data Analysis

Key Observations:
- Salary is capped at 120,000
- Salary varies significantly by country
- Salary varies by college tier
- Weak linear relationship with most numeric predictors

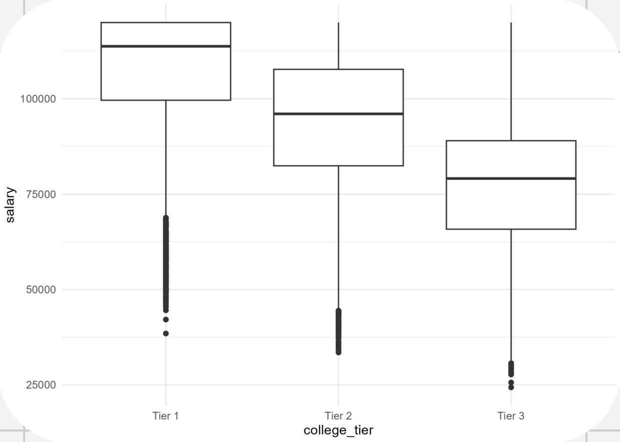
Distribution of Salary for Placed students



Salary by Country



Salary by College Tier



Exploratory Data Analysis

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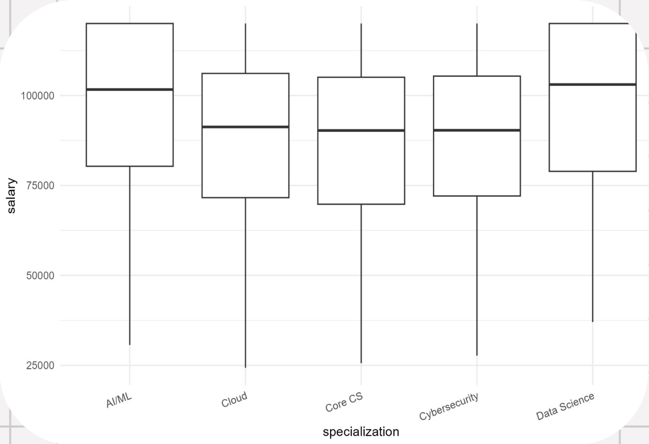
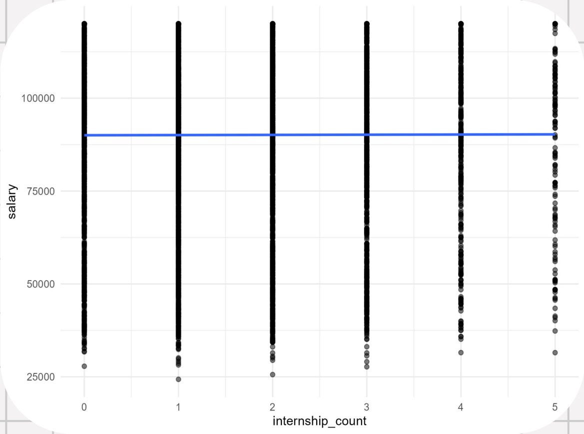
CGPA vs Salary by Country

Salary by Specialization



Salary vs Internship Count



Variable Selection

Full Model

```
Call:
lm(formula = salary ~ cgpa + backlogs + college_tier + country +
    university_ranking_band + specialization + industry + internship_count +
    internship_quality_score + aptitude_score + communication_score,
    data = df_placed)

Residuals:
     Min       1Q   Median       3Q      Max
-19157.9  -3876.4    152.7   4089.1  17379.0

Coefficients:
                              Estimate Std. Error  t value Pr(>|t|)
(Intercept)                  81523.494    978.560   83.310   <2e-16 ***
cgpa                          4340.890    100.664   43.123   <2e-16 ***
backlogs                        17.006     78.669    0.216    0.829
college_tierTier 2          -12004.327    176.789  -67.902   <2e-16 ***
college_tierTier 3          -27527.263    189.474 -145.283   <2e-16 ***
countryGermany               -3584.236    232.793  -15.397   <2e-16 ***
countryIndia                -45845.539    231.898 -197.697   <2e-16 ***
countryUK                     3396.546    235.374   14.430   <2e-16 ***
countryUSA                   14069.448    231.019   60.902   <2e-16 ***
university_ranking_band300+  -2222.131    167.841  -13.239   <2e-16 ***
university_ranking_bandTop 100 2689.660   191.578   14.039   <2e-16 ***
specializationCloud          -9778.895    231.497  -42.242   <2e-16 ***
specializationCore CS       -10149.481    234.114  -43.353   <2e-16 ***
specializationCybersecurity  -9908.113    232.245  -42.662   <2e-16 ***
specializationData Science    -263.809    232.175   -1.136    0.256
industryFinance               3676.700    249.692   14.725   <2e-16 ***
industryHealthcare             287.737    248.614    1.157    0.247
industryManufacturing          358.263    258.169    1.388    0.165
industryOther                   90.386    259.061    0.349    0.727
industryTech                  6742.662    246.701   27.331   <2e-16 ***
internship_count                 6.798     59.251    0.115    0.909
internship_quality_score        32.267     50.061    0.645    0.519
aptitude_score                   7.662      5.072    1.511    0.131
communication_score              3.829      4.998    0.766    0.444
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5752 on 6129 degrees of freedom
Multiple R-squared:  0.9468, Adjusted R-squared:  0.9466
F-statistic:  4746 on 23 and 6129 DF,  p-value: < 2.2e-16
```

VIF

```
                             GVIF
cgpa                     1.059251
backlogs                 1.052334
college_tier             1.007133
country                  1.013564
university_ranking_band  1.013531
specialization           1.013047
industry                 1.024403
internship_count         1.011771
internship_quality_score 1.021103
aptitude_score           1.004961
communication_score      1.005536
```

All VIF scores were below 5, indicating no severe multicollinearity. We can proceed without removing any predictors for this reason.

Single-Predictor Regression Screening

- Ran simple linear regression on each predictor
- Most predictors have low R^2 individually
- Indicates weak standalone predictive power
- Categorical variables (country, college tier) show stronger effects
- Any value labeled 0 is <0.001

Removed Variables:
- backlogs
- internship count
- internship quality
- communication score

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Variable Selection

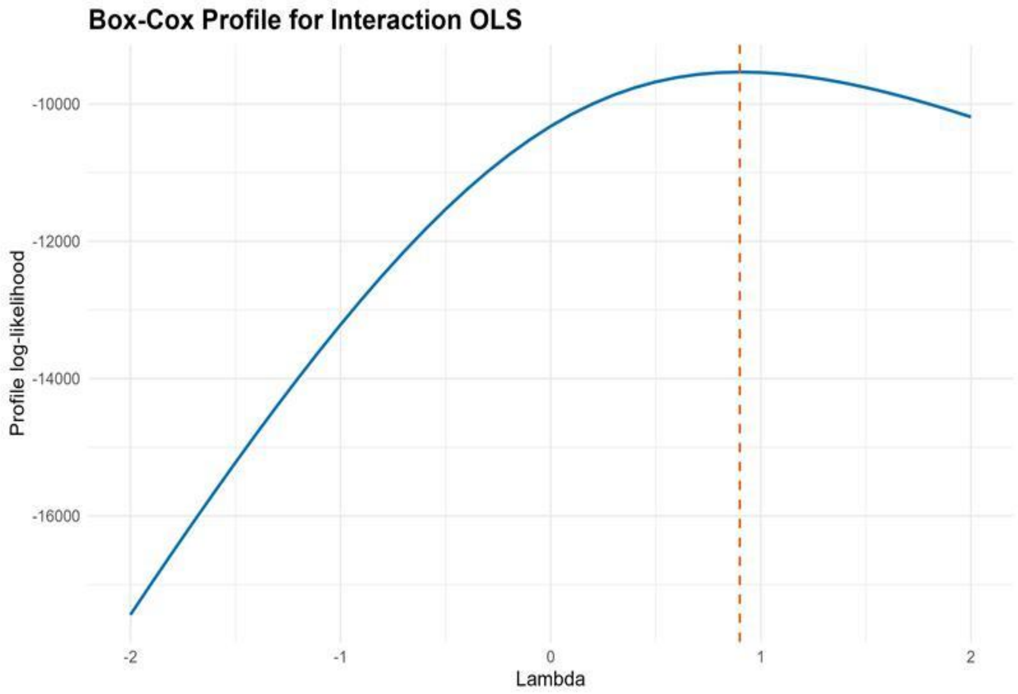
Selected model performs essentially the same as full model, but with fewer variables and slightly better AIC / adjusted R^2.

Removed as low-value after controlling for others: backlogs, internship_count, communication_score, internship_quality_score

Full vs Selected Model

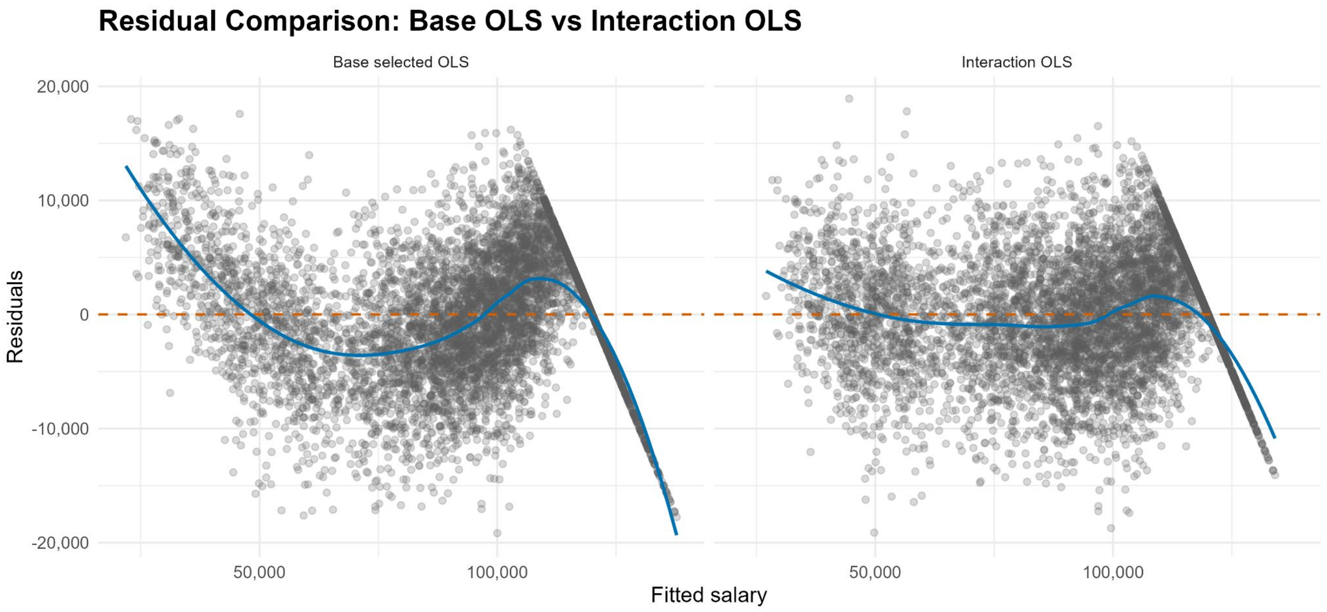
Model	AIC	Adj. R^2	Terms
Full	124024.16	0.946642	11
Selected	124017.23	0.946667	7

Model Transformation



Best λ	0.9
Interpretation	Box-Cox suggests little or no transformation is needed
Note	Arcsine transformation was not tested because salary is not a 0–1 proportion variable

Model Selection

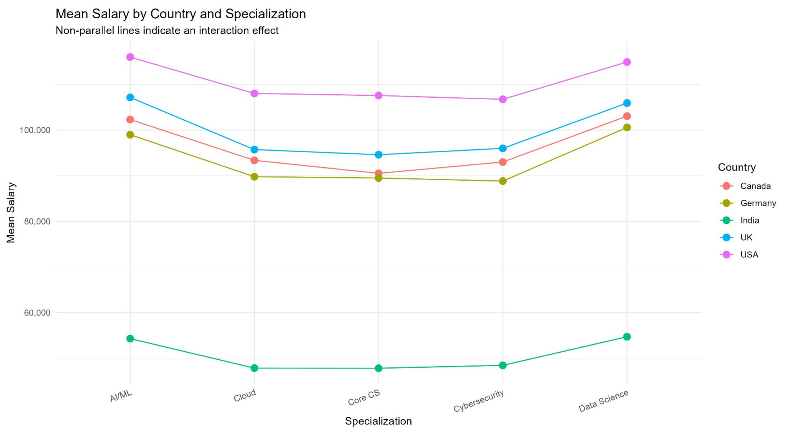


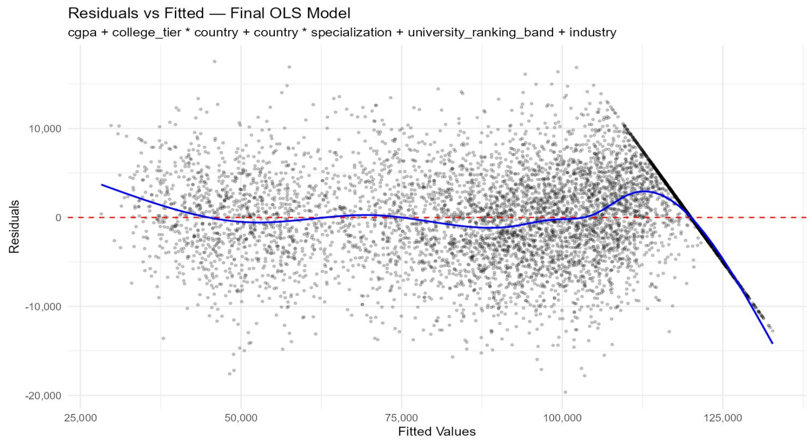
Base selected OLS model: salary ~ cgpa + aptitude_score + college_tier + country + university_ranking_band + specialization + industry

Interaction OLS model: salary ~ cgpa + aptitude_score + college_tier * country + university_ranking_band + specialization + industry

Model Selection

Model Building



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Analysis of Variance Table

Model 1: salary ~ cgpa + aptitude_score + college_tier * country + university_ranking_band +
    specialization + industry
Model 2: salary ~ cgpa + aptitude_score + college_tier * country + country *
    specialization + university_ranking_band + industry
  Res.Df        RSS Df  Sum of Sq      F    Pr(>F)
1   6125 1.6411e+11
2   6109 1.5596e+11 16 8158132621 19.973 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

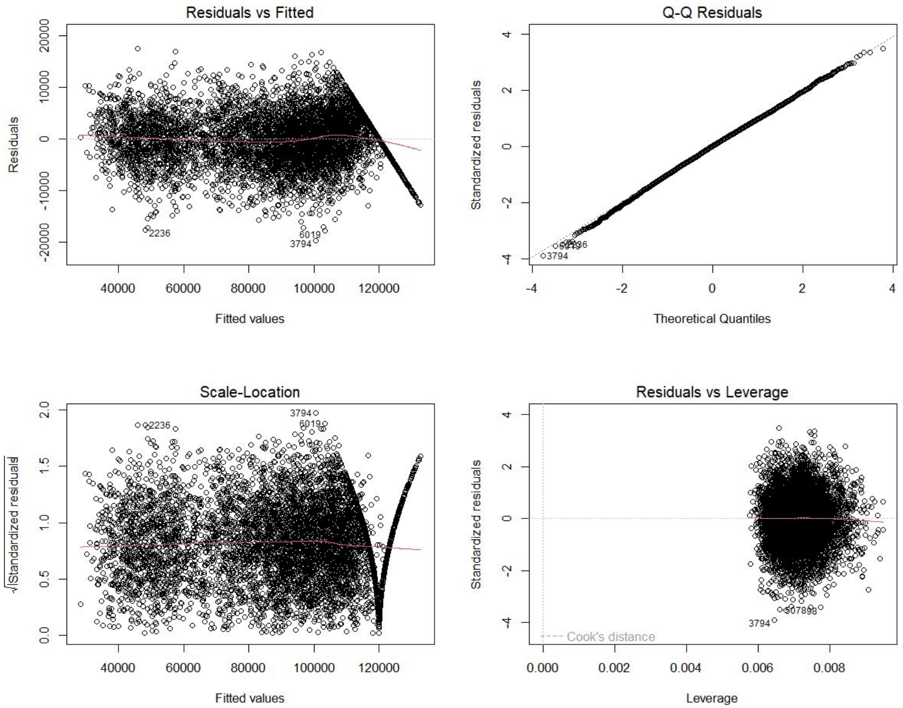
Multiple Regression and Model Selection

- Fit full model with all predictors
- Used stepwise AIC for model selection
- Compared models using:
    - Adjusted R^2
    - AIC

Model	Predictors Included	AIC	Adj R^2	Notes
Full Model	All predictors	124024.16	0.946642	Baseline comparison
Stepwise Model	cgpa + aptitude_score + college_tier * country + university_ranking_band + specialization + industry	124017.23	0.946667	Candidate final model
Final Model	cgpa + aptitude_score + college_tier * country + country * specialization + university_ranking_band + industry	122448.65	0.958829	Selected after diagnostics

Model Diagnostics & Validation

Final Model

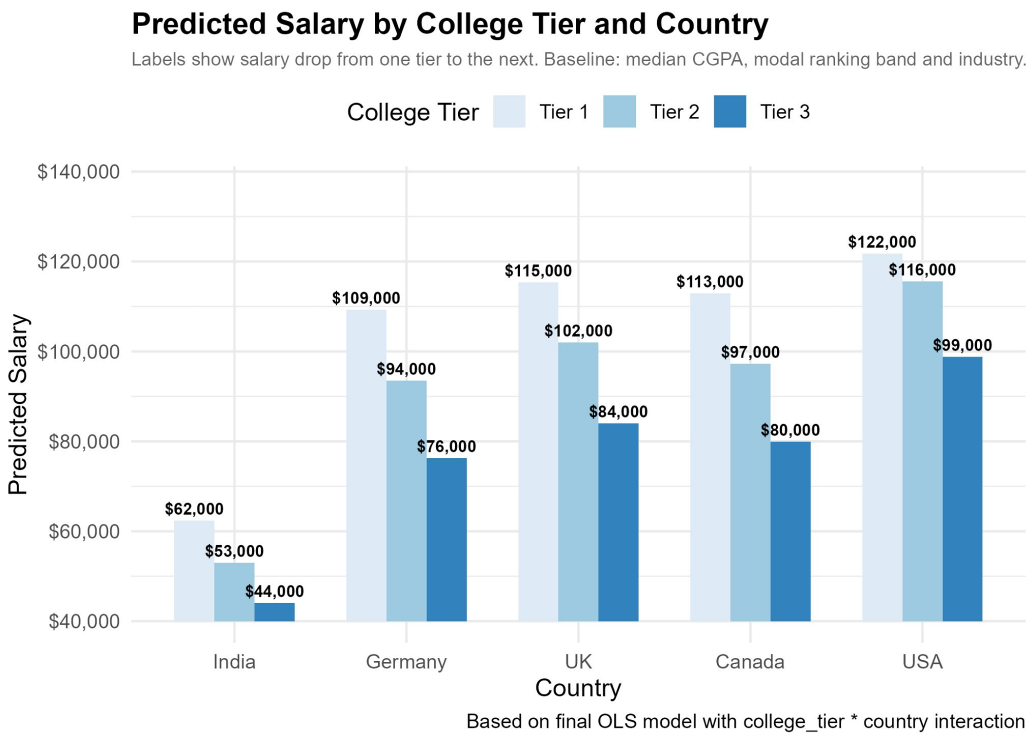


Final Model

model	n	residual_df	r_squared	adjusted_r_squared	residual_standard_error
final_ols_country_tier_country_specialization	6153	6109	0.9591163642	0.9588285927	5052.5971

Model component	Coefficients
Intercept	1
CGPA + aptitude score	2
College tier main effects	2
Country main effects	4
College tier × country	8
Specialization main effects	4
Country × specialization	16
University ranking band	2
Industry	5
Total	**44**

Final Model Interpretation



Conclusion

Diagnostics

- Residual vs fitted plot
- QQ plot
- Checked for constant variance
- Checked influential points (Cook's D, leverage)

Conclusion

- Strongest predictor: Country
- Salary influenced more by categorical factors than numeric ones
- Salary cap at 120,000 affects results

Future Work

- Try log(salary)
- Explore interaction effects
- Analyze by country separately

Questions ?

Good Luck on Finals!